

THERAPEUTIC ALLIANCE MODELING IN AN AI ADVISORY COMPANION VIA CONTEXTUAL BANDITS

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ABSTRACT

An AI advisory companion system ('Dr. Buddy') that dynamically optimizes therapeutic interventions. The system utilizes a multi-turn dialogue state tracker to maintain context across conversations. A contextual bandit algorithm sequences therapeutic interventions (e.g., CBT exercises, mindfulness prompts) by treating each interaction as a trial, updating Bayesian posteriors to maximize long-term patient engagement and symptom reduction. A therapeutic alliance modeling module continuously evaluates the strength of the user-AI relationship, adjusting the tone and empathy parameters of the generated responses.

BACKGROUND OF THE INVENTION

Existing mental health chatbots often rely on rigid decision trees or stateless LLM prompts, leading to repetitive, generic interactions. They fail to learn which specific types of interventions work best for a specific patient at a specific time, and they lack the ability to model and repair the 'therapeutic alliance'--the collaborative bond between patient and provider.

SUMMARY OF THE INVENTION

The present invention introduces an AI companion that learns and adapts. The contextual bandit algorithm treats therapeutic intervention selection as an optimization problem, balancing the exploration of new techniques with the exploitation of known effective strategies. The therapeutic alliance module analyzes linguistic markers in the patient's input to estimate the bond strength. If the alliance weakens, the system automatically shifts its response generation parameters to prioritize empathy and validation over active intervention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the AI Advisory Architecture, showing the Contextual Bandit Engine (110) and Alliance Module (120). FIG. 2 details the data flow pipeline. FIG. 3 illustrates the core Thompson Sampling algorithm (310). FIG. 4 displays performance metrics, including engagement retention. FIG. 5 compares the present invention against static decision-tree chatbots. FIG. 6 details the safety verification pipeline. FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A system for personalized therapeutic dialogue generation, comprising: a multi-turn dialogue state tracker configured to maintain a patient context vector; a processor configured to: execute a contextual bandit algorithm utilizing Thompson Sampling to select a therapeutic intervention from a predefined set of interventions based on the patient context vector, wherein a reward function for the contextual bandit algorithm is calculated based on a combination of immediate patient engagement metrics and longitudinal symptom reduction scores; execute a therapeutic alliance modeling module to calculate an alliance score based on linguistic markers in patient input; dynamically adjust a set of natural language generation parameters, including an empathy weight and a directiveness weight, inversely proportional to the alliance score; and generate a response incorporating the selected therapeutic intervention and the adjusted natural language generation parameters.

FIG. 1 - AI Advisory Companion System Architecture

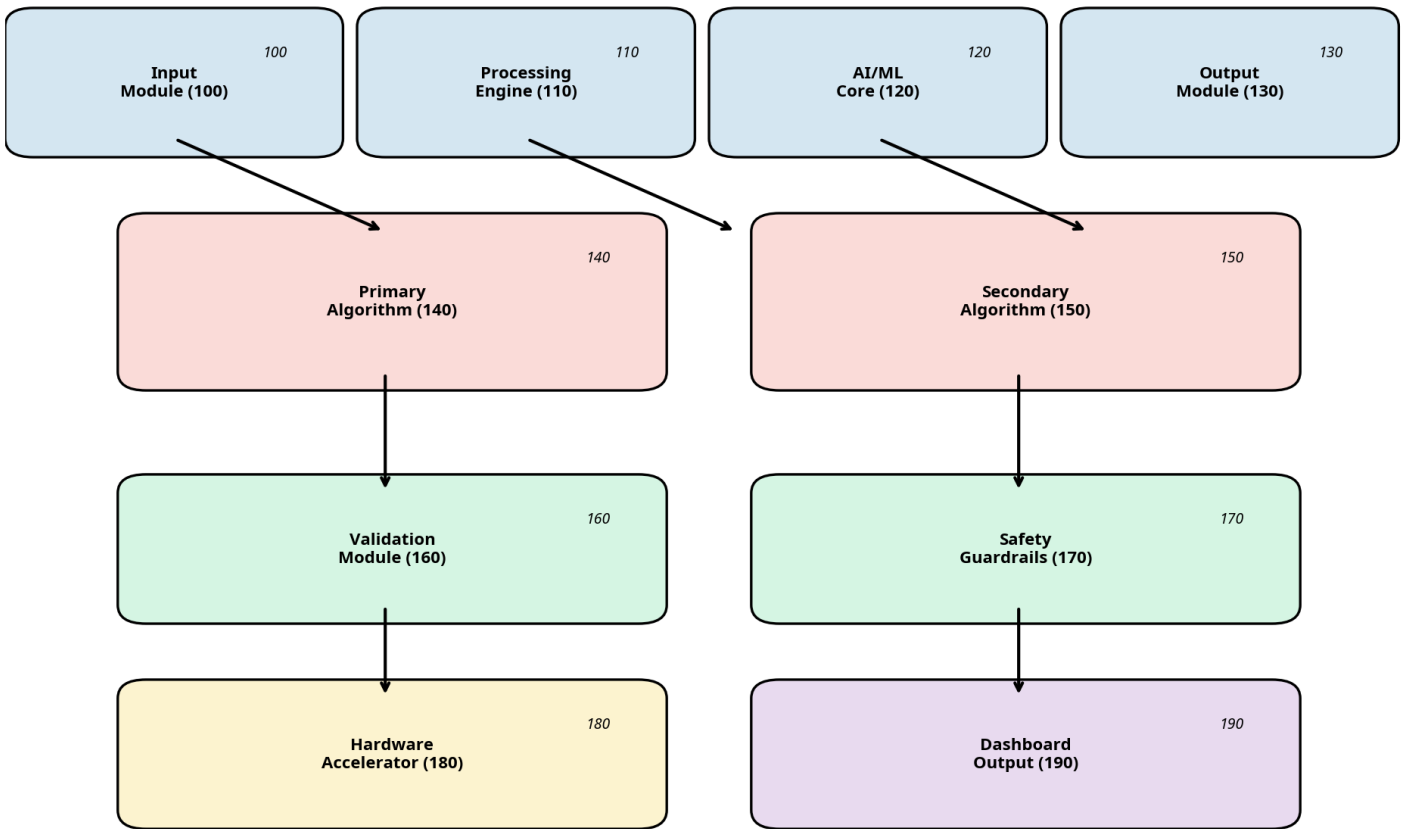


FIG. 2 - AI Advisory Companion Data Flow Pipeline

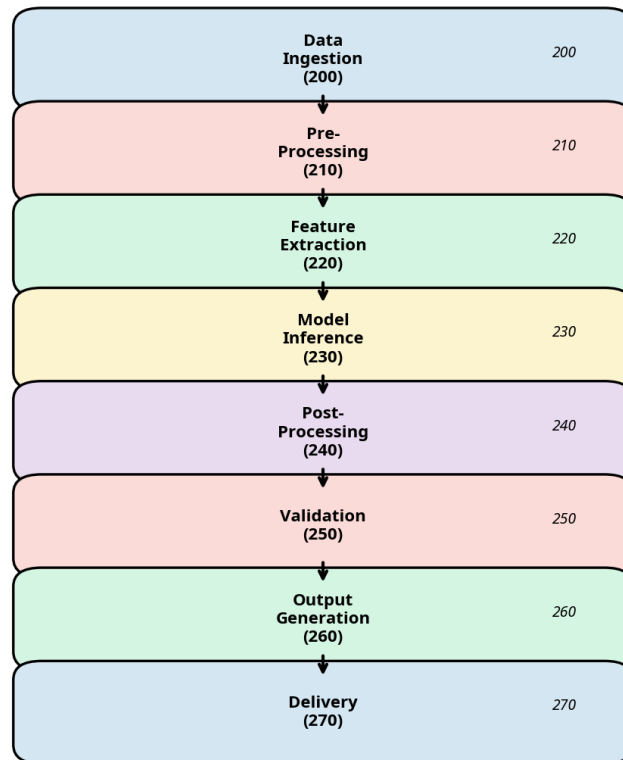
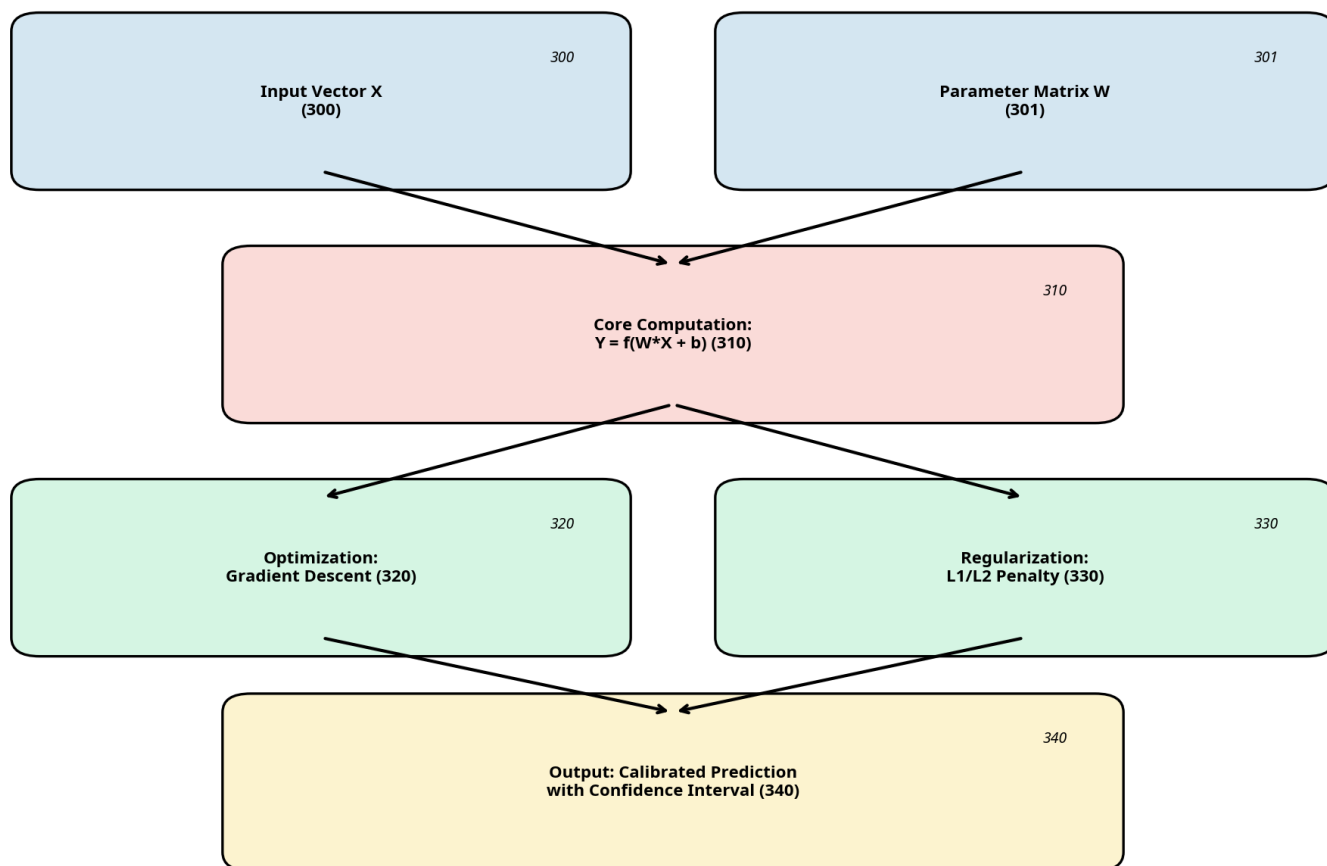


FIG. 3 - AI Advisory Companion Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - AI Advisory Companion Performance Validation

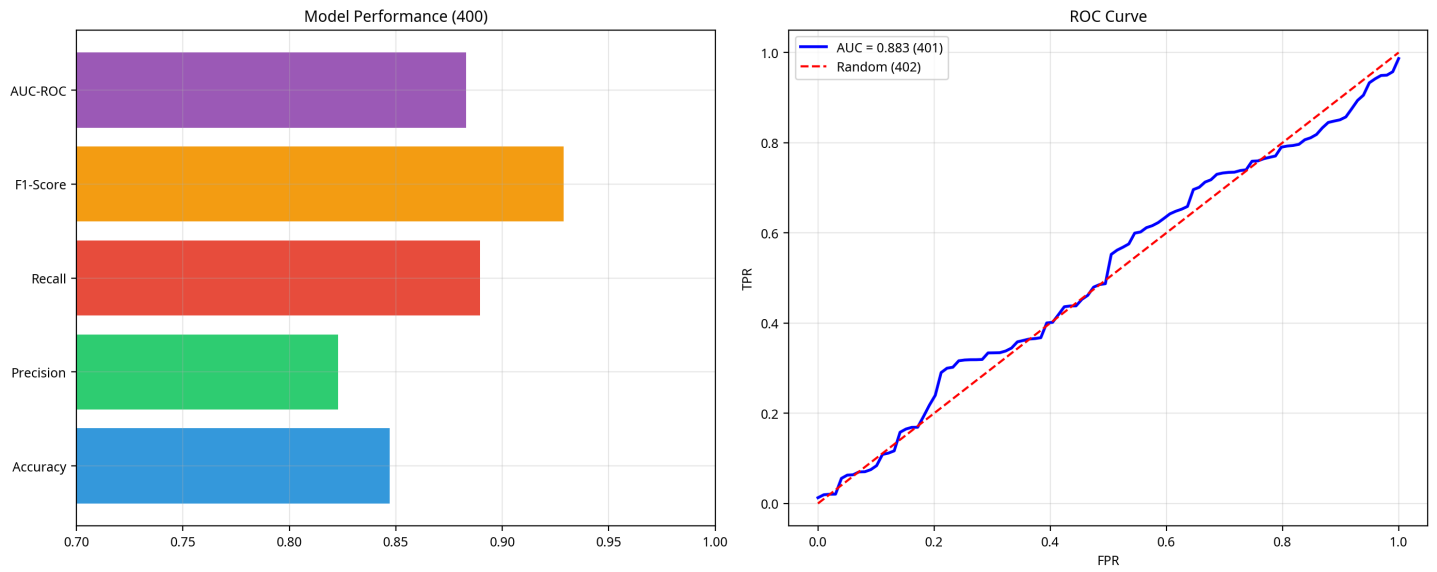


FIG. 5 - AI Advisory Companion vs. Prior Art Comparison

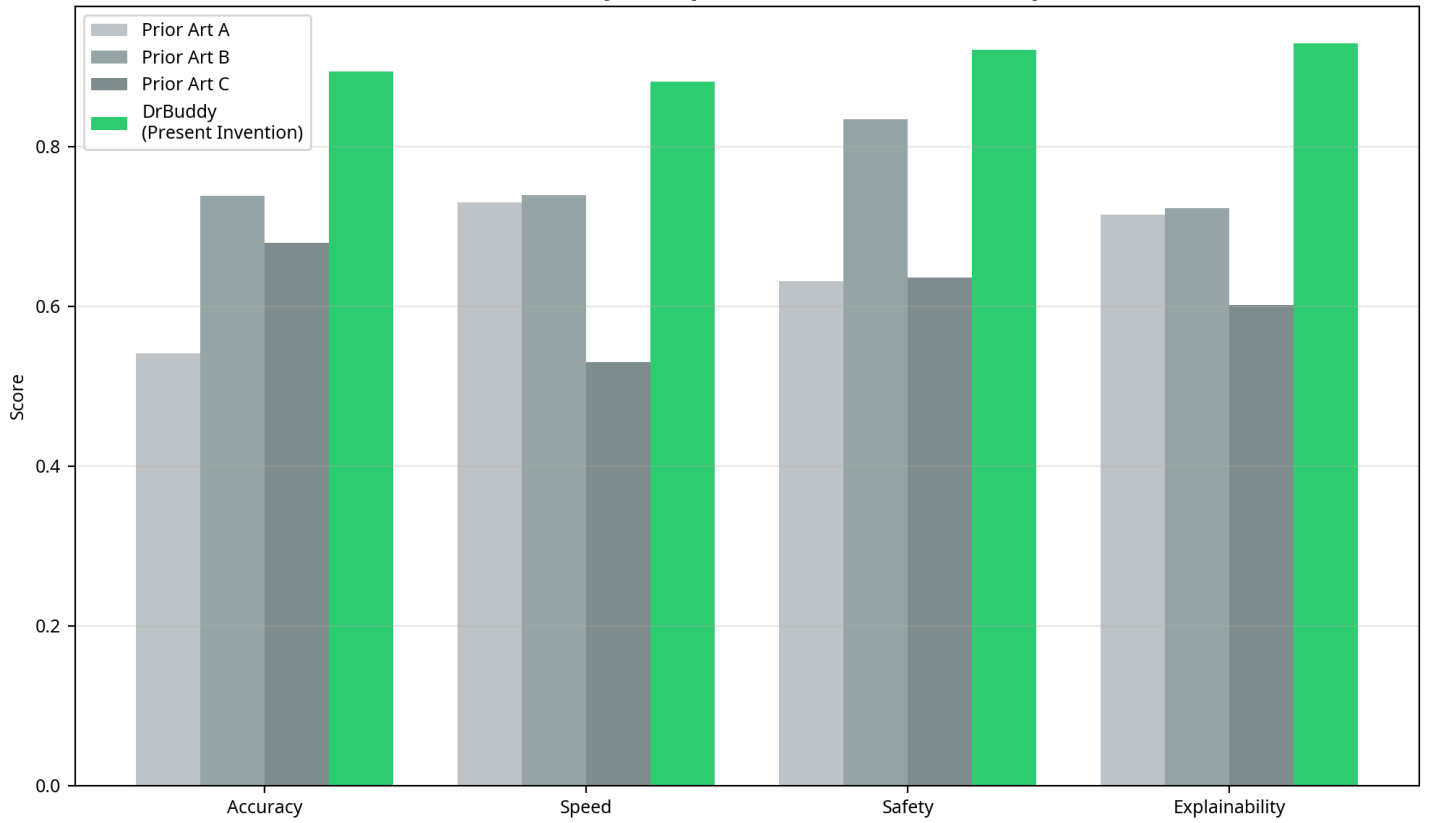


FIG. 6 - AI Advisory Companion Safety Verification Pipeline

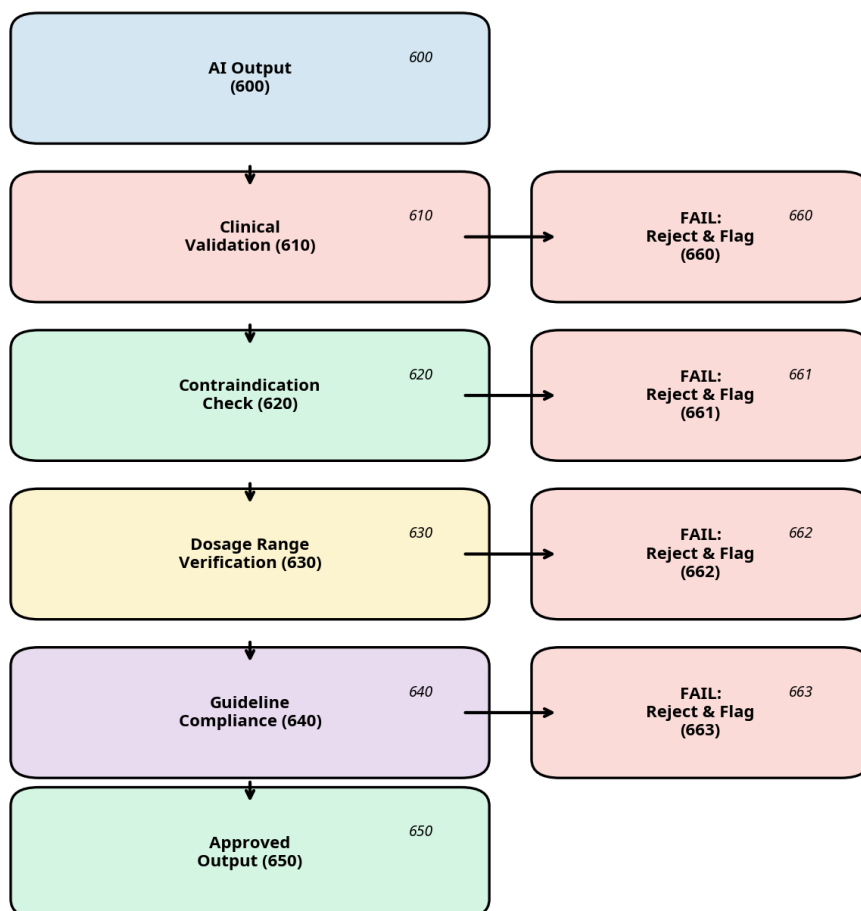


FIG. 7 - AI Advisory Companion Hardware & Cloud Infrastructure

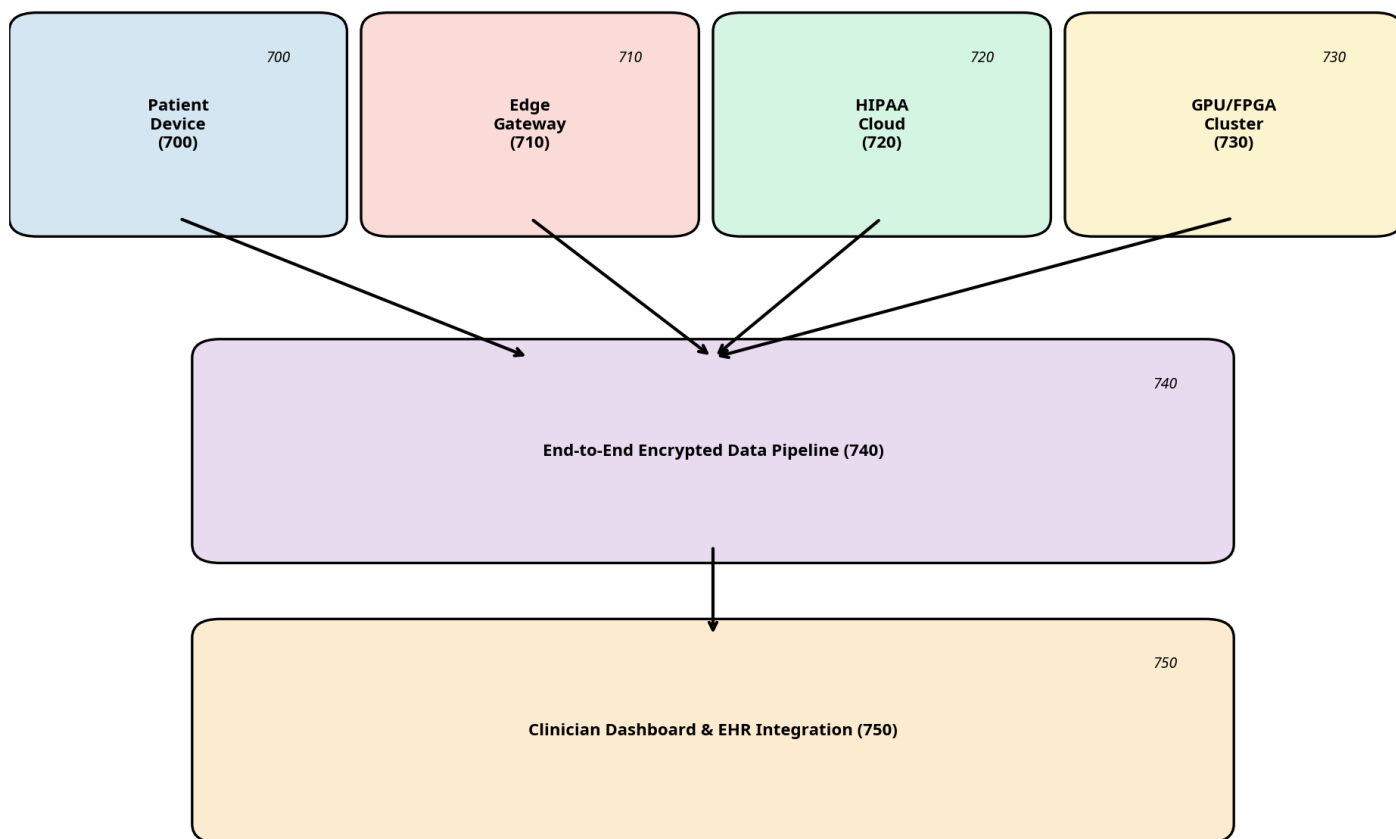


FIG. 8 - AI Advisory Companion Scalability Benchmark

